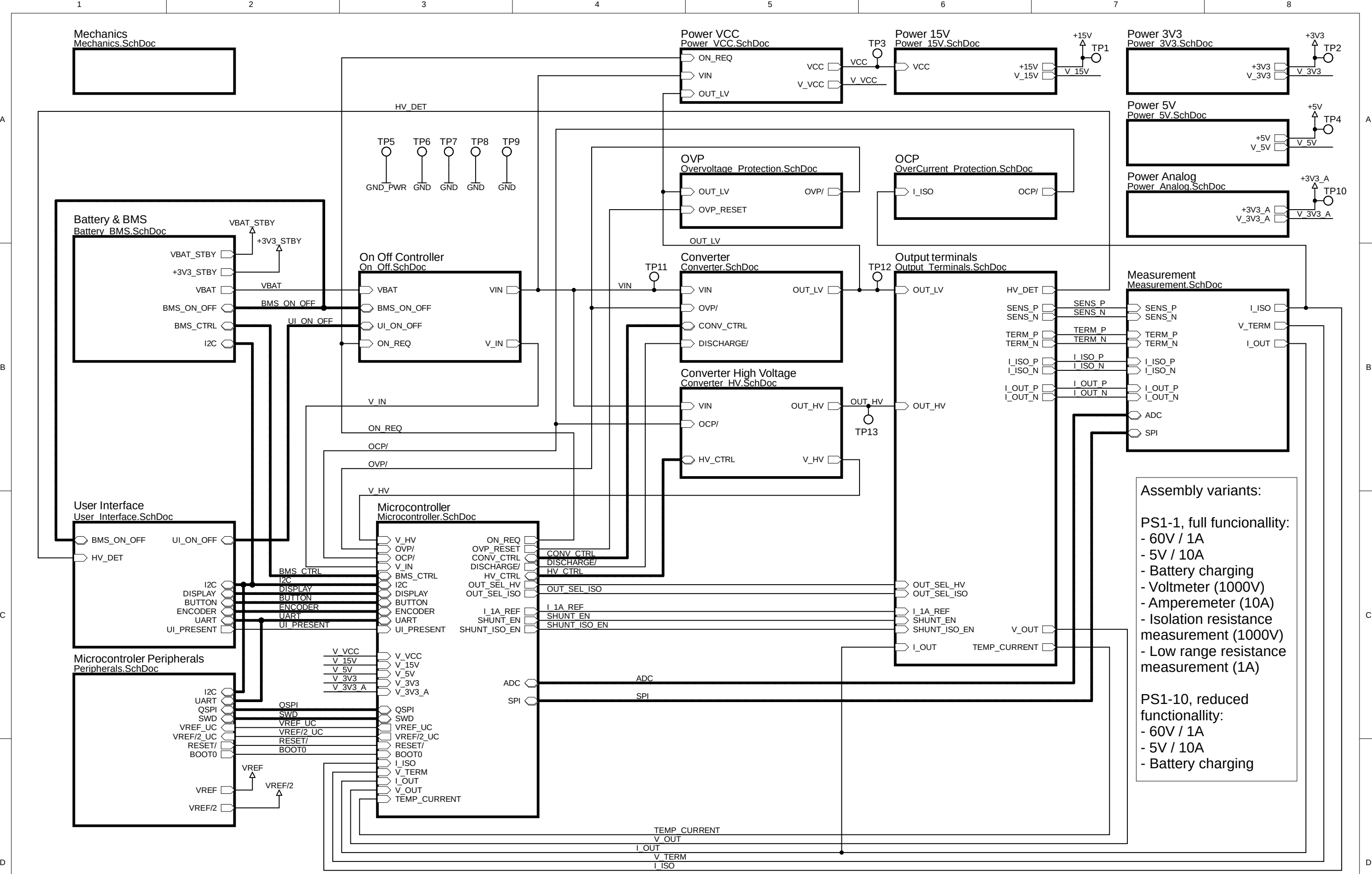




Assembly variants:

PS1-1, full functionality:

- 60V / 1A
- 5V / 10A
- Battery charging
- Voltmeter (1000V)
- Amperemeter (10A)
- Isolation resistance measurement (1000V)
- Low range resistance measurement (1A)

PS1-10, reduced functionality:

- 60V / 1A
- 5V / 10A
- Battery charging

 M3
SERIAL 2D

 M6
QR PROJ URL

ST1-1B

LOGO

LOGO

M1
PCB NAME

M4
LOGO WHE

M7
LOGO FSG

 M2
Tooling JLC

 M5
Tooling JLC

FM1

FM2

FM3

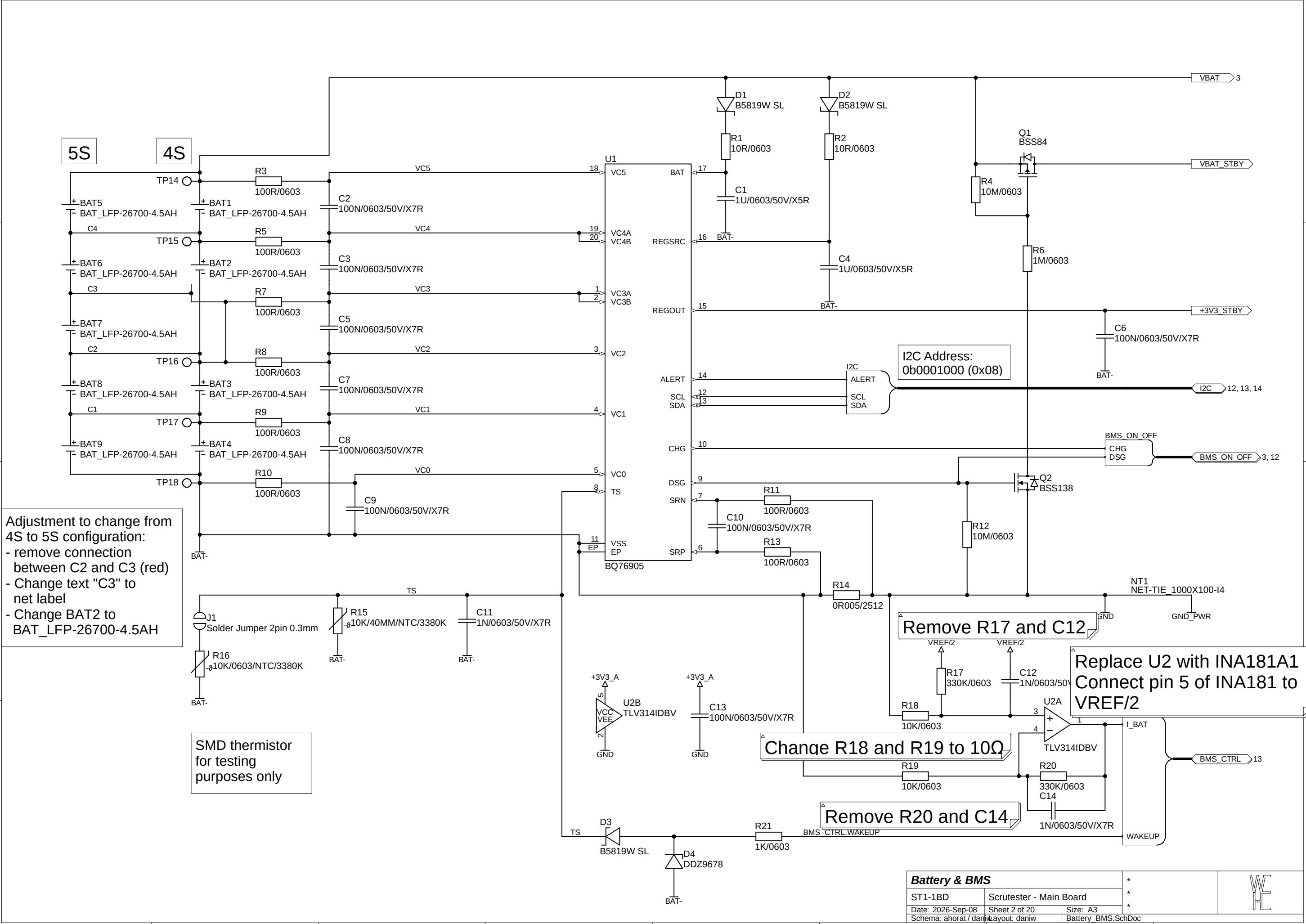
FM4

FM5

FM6

Top			*	
ST1-1BD	Scrutester - Main Board		*	
Date: 2026-Sep-08	Sheet 1 of 20	Size: A3	*	
Schema: ahorat / daniw	Layout: daniw	ST1-1-Top.SchDoc		





Adjustment to change from 4S to 5S configuration:

- remove connection between C2 and C3 (red)
- Change text "C3" to net label
- Change BAT2 to BAT_LFP-26700-4.5AH

SMD thermistor for testing purposes only

Remove R17 and C12

Replace U2 with INA181A1
Connect pin 5 of INA181 to VREF/2

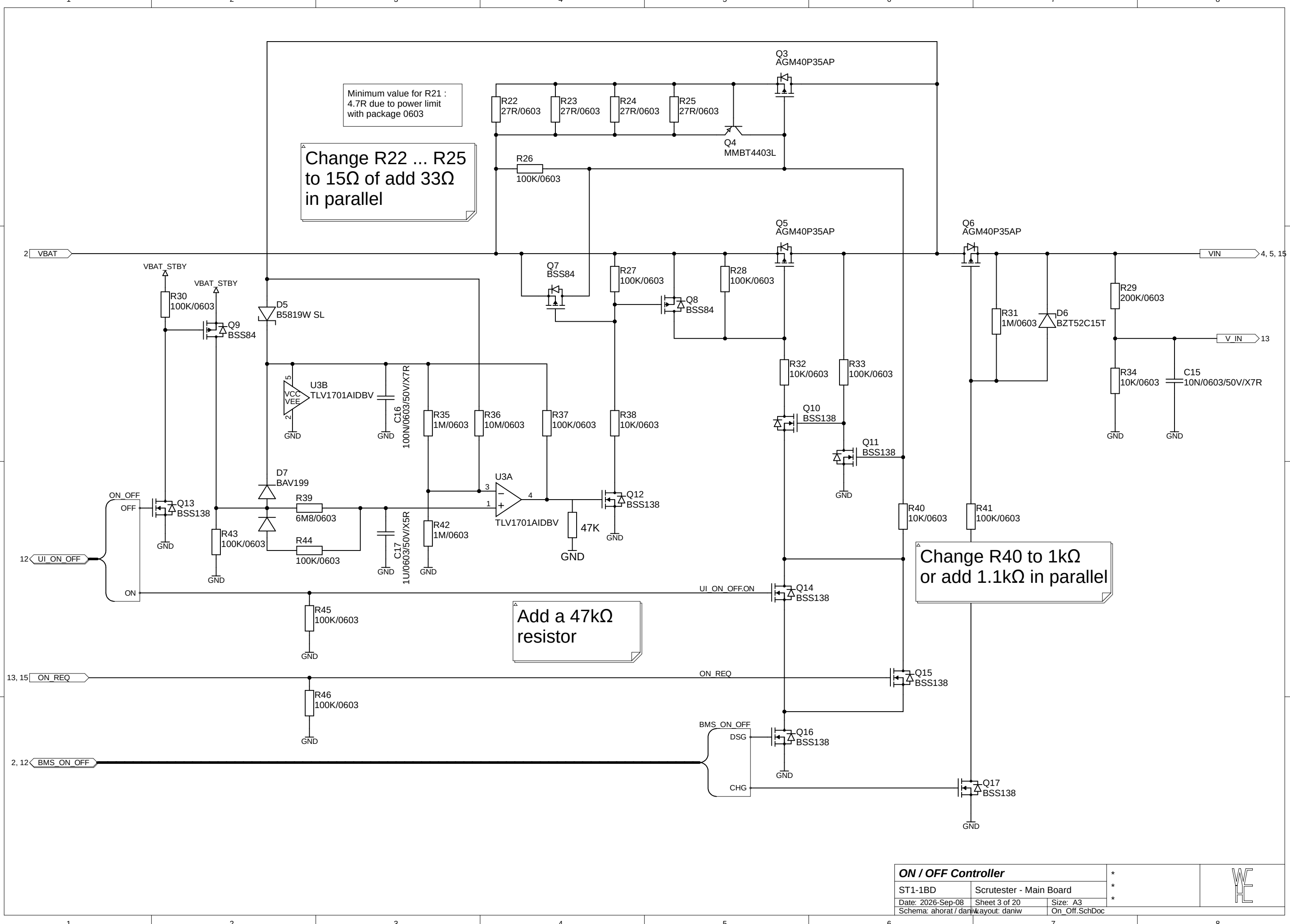
Change R18 and R19 to 10Ω

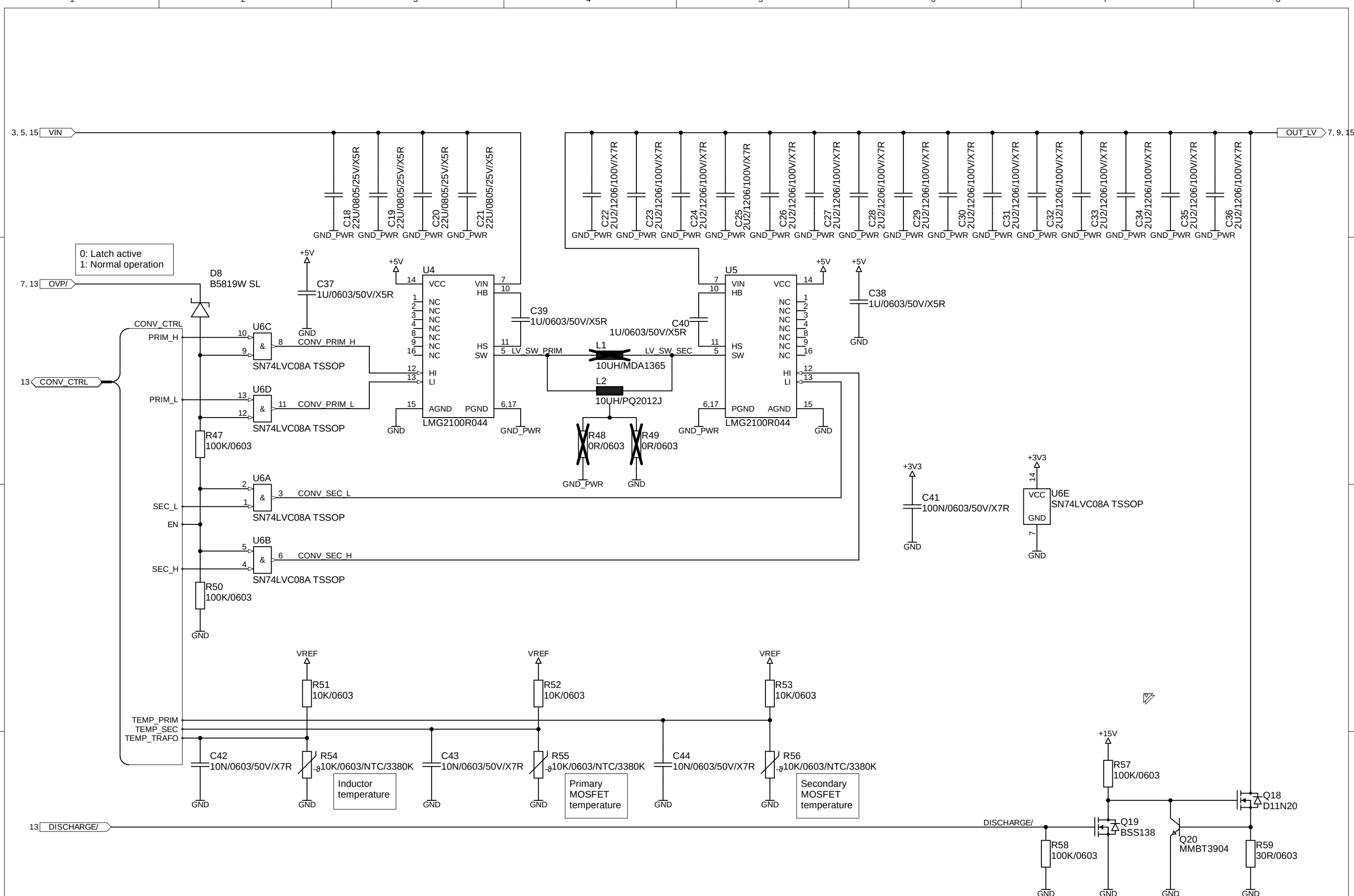
Remove R20 and C14

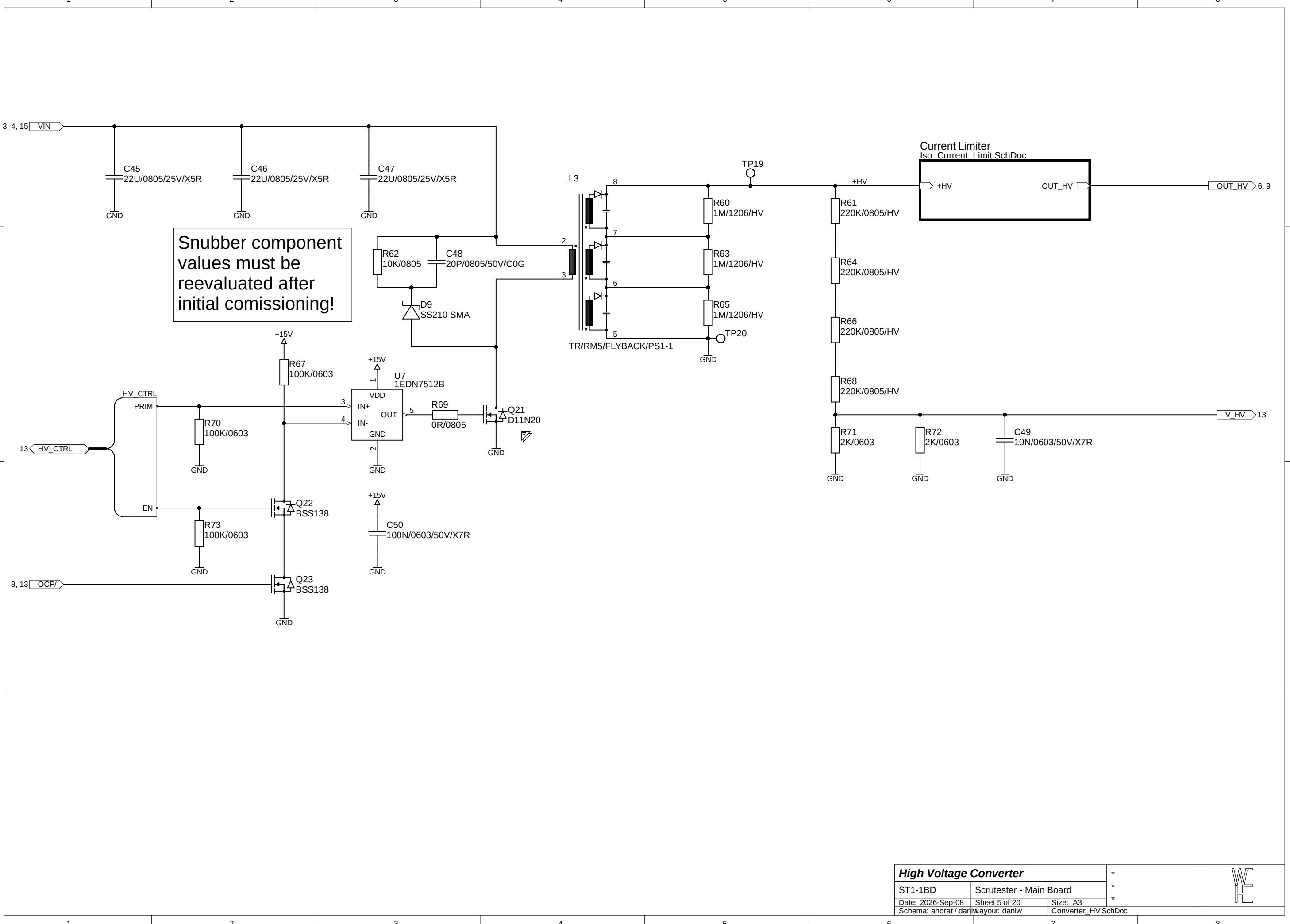
Battery & BMS

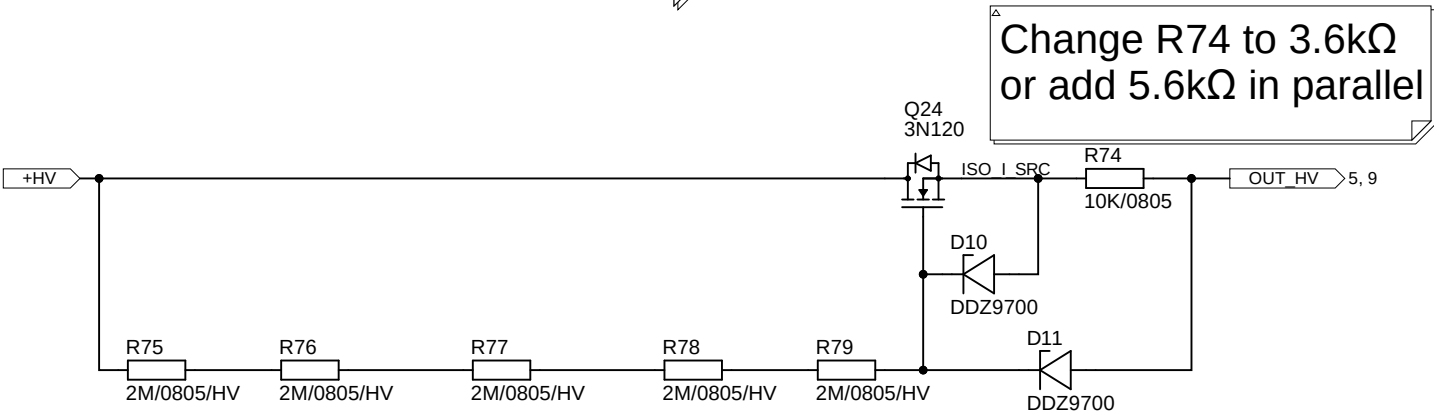
ST1-1BD	Scrutester - Main Board	
Date: 2026-Sep-08	Sheet 2 of 20	Size: A3
Schema: ahorat / danw	Layout: danw	Battery_BMS.SchDoc



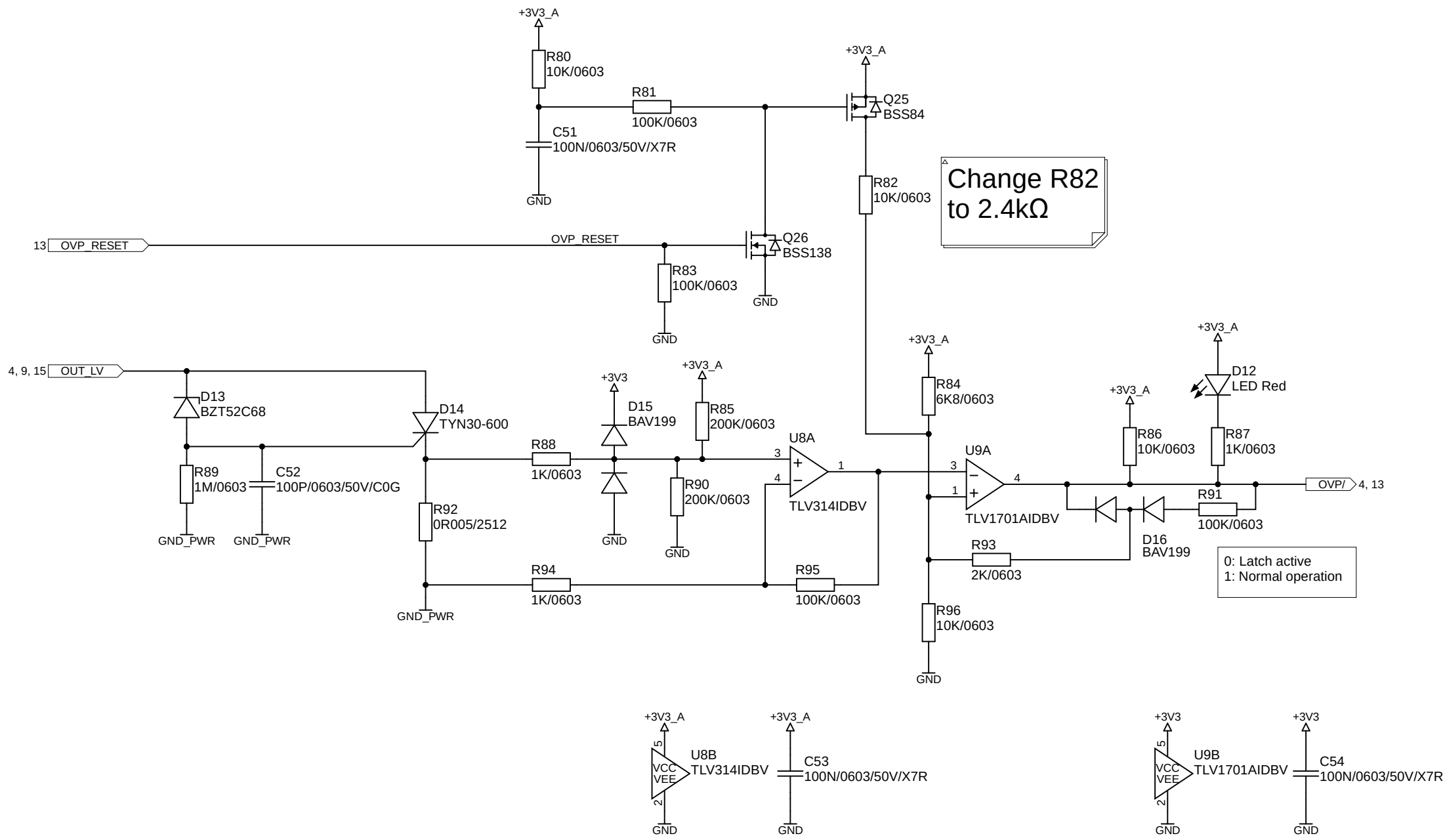


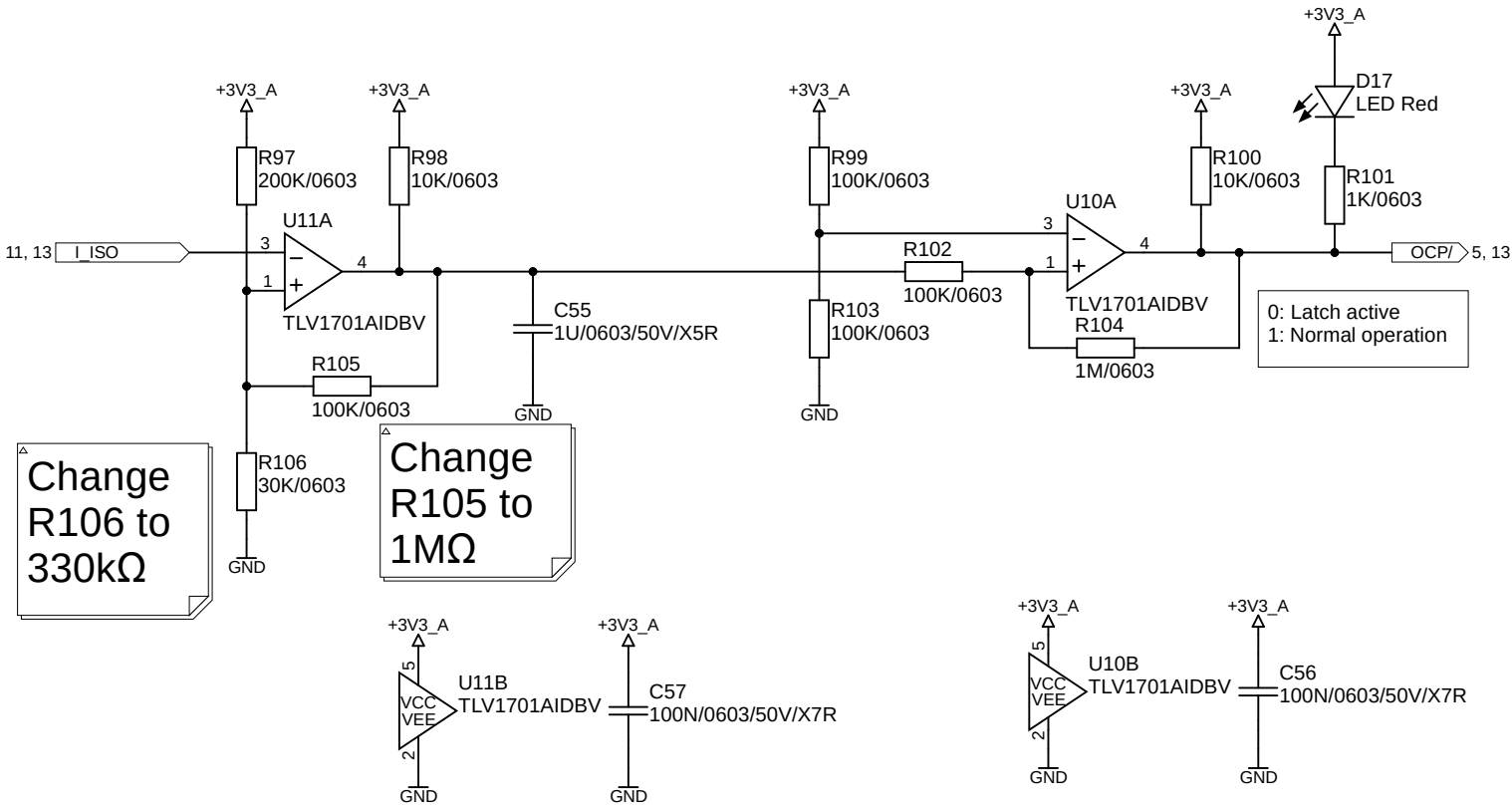


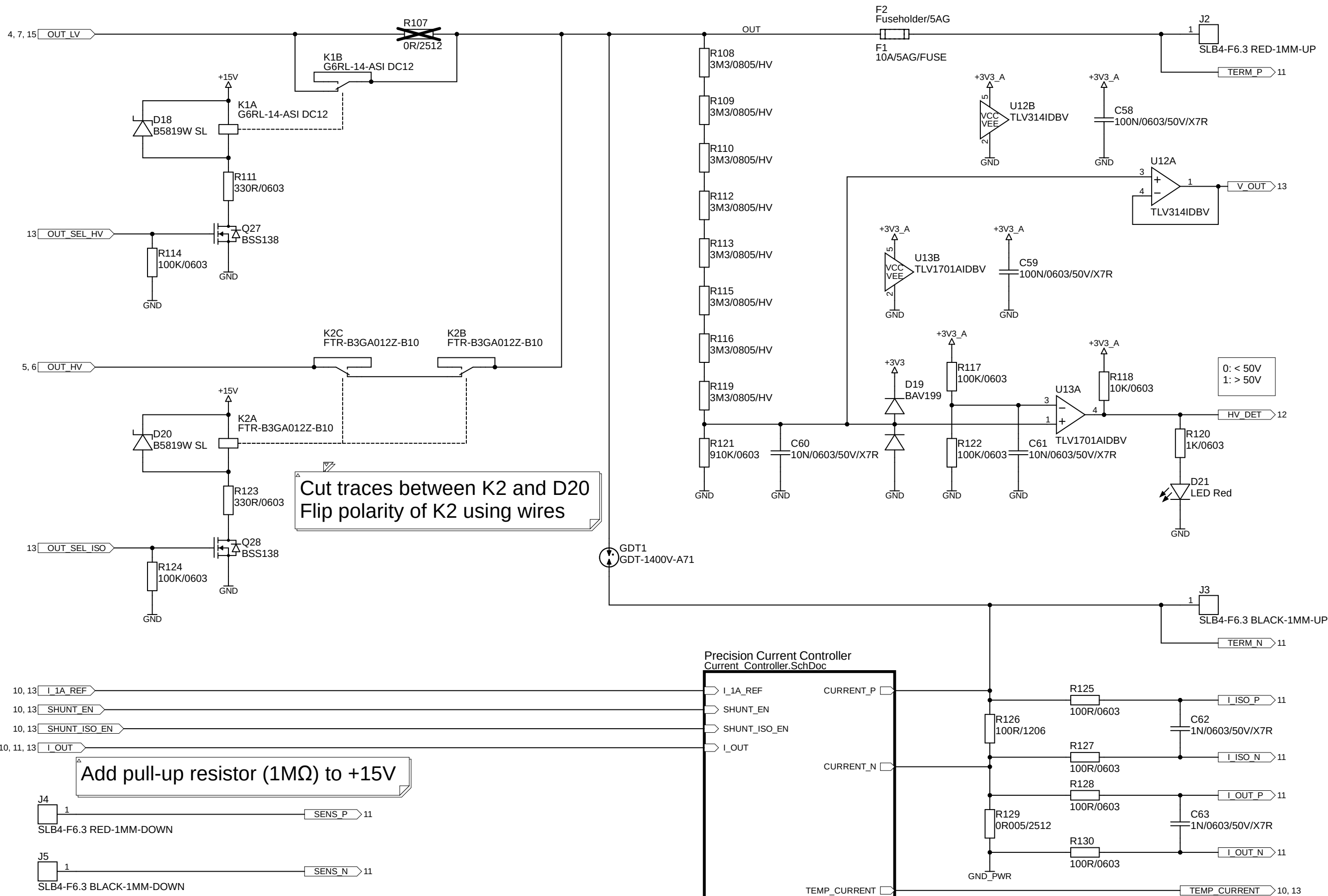




Current Limiter			* * * WE
ST1-1BD	Scrutester - Main Board		
Date: 2026-Sep-08	Sheet 6 of 20	Size: A3	
Schema: ahorat / daniw		Layout: daniw	
Iso_Current_Limit.SchDoc			



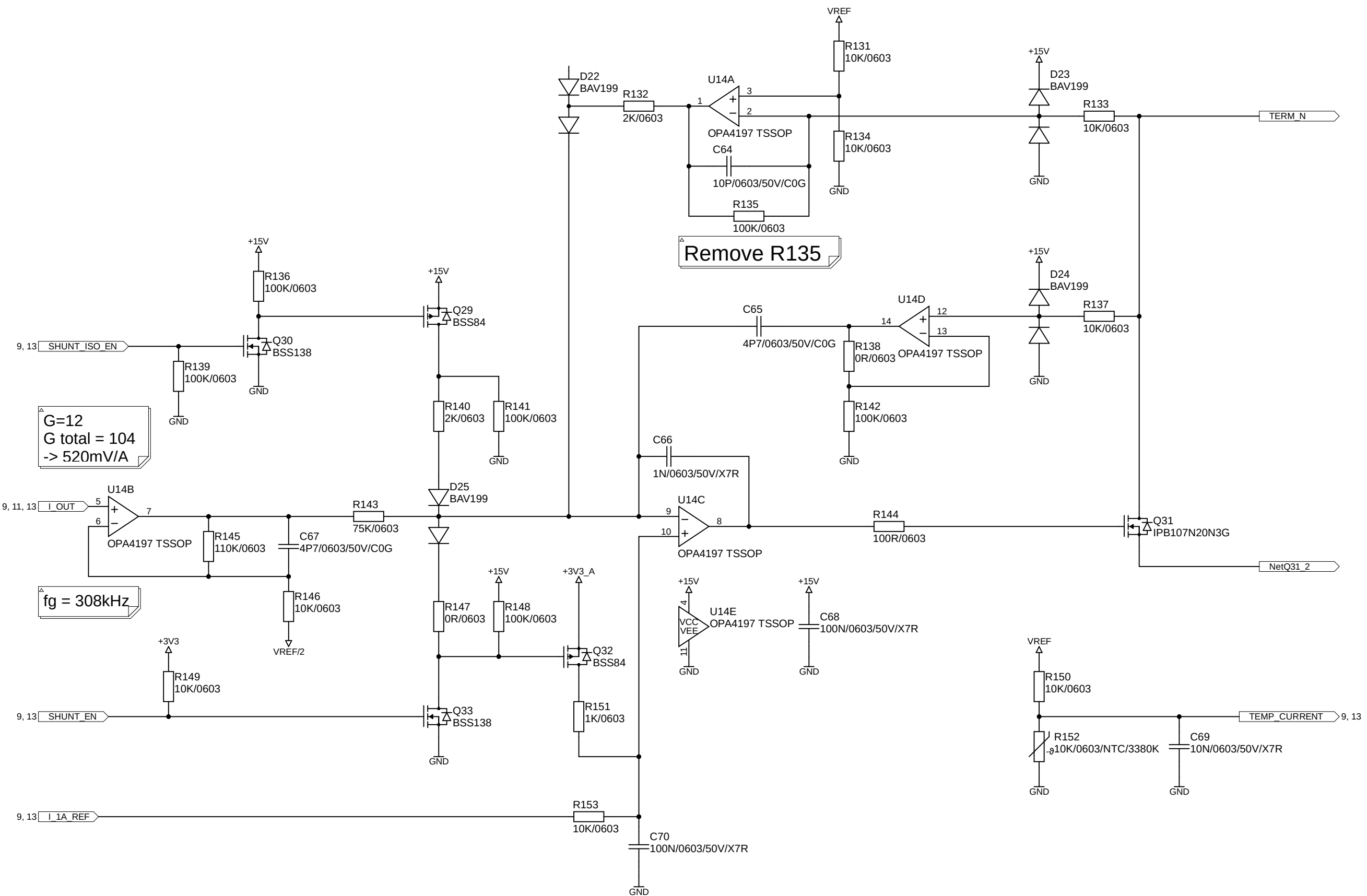


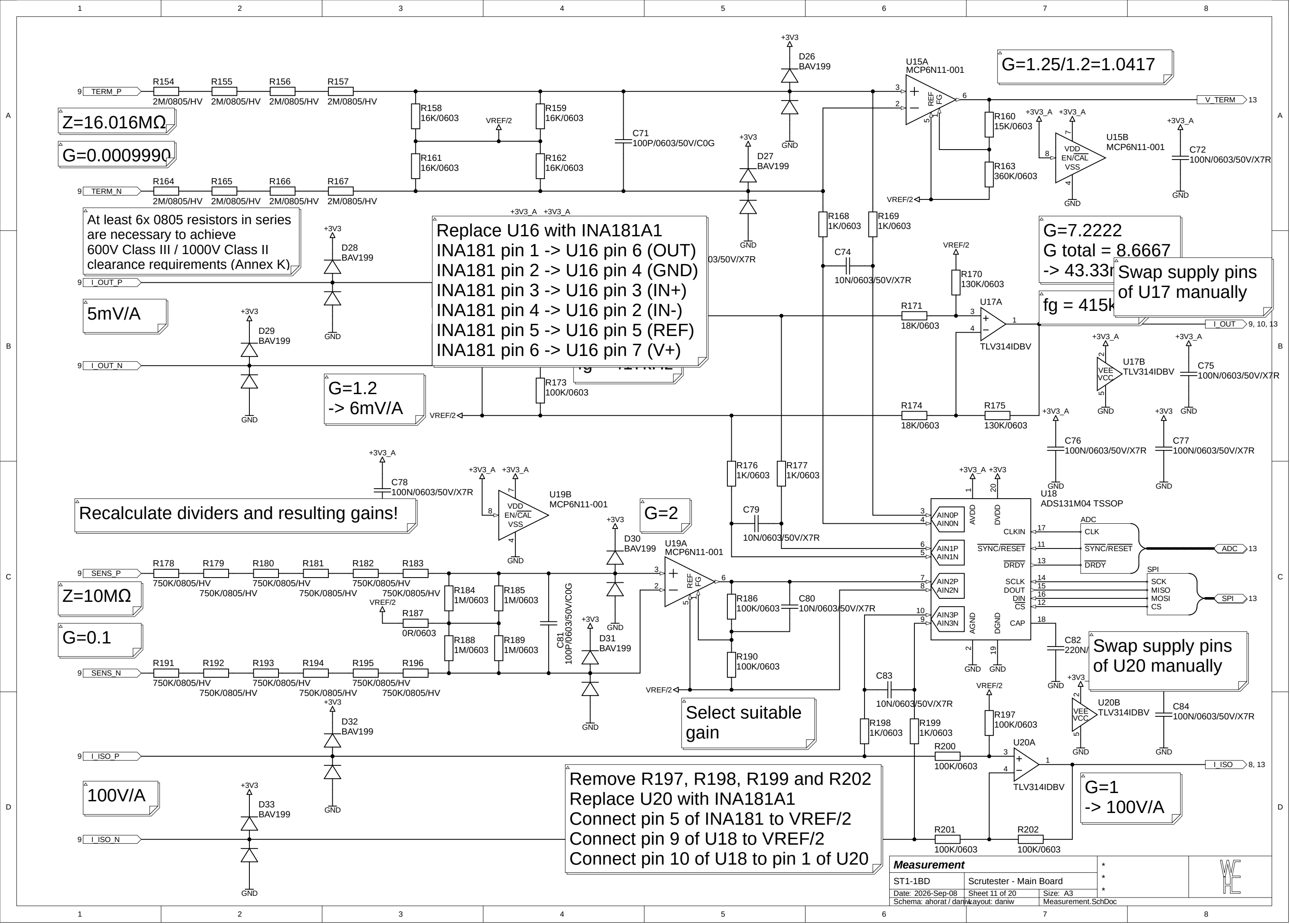


Output Terminals & Voltage detection

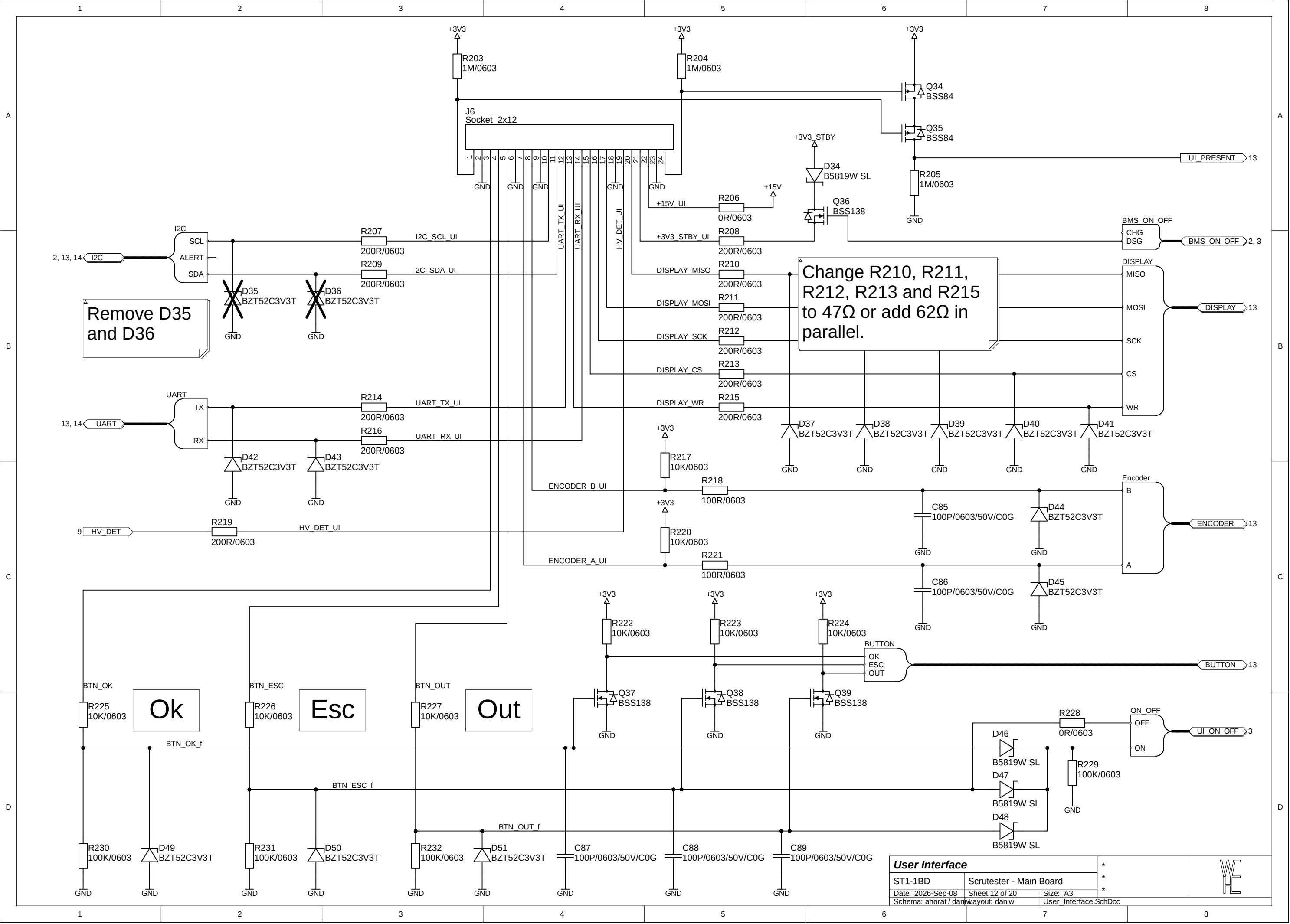
ST1-1BD	Scrutester - Main Board	*
Date: 2026-Sep-08	Sheet 9 of 20	*
Schema: ahorat / danil	Layout: danilw	*
	Output_Terminals.SchDoc	

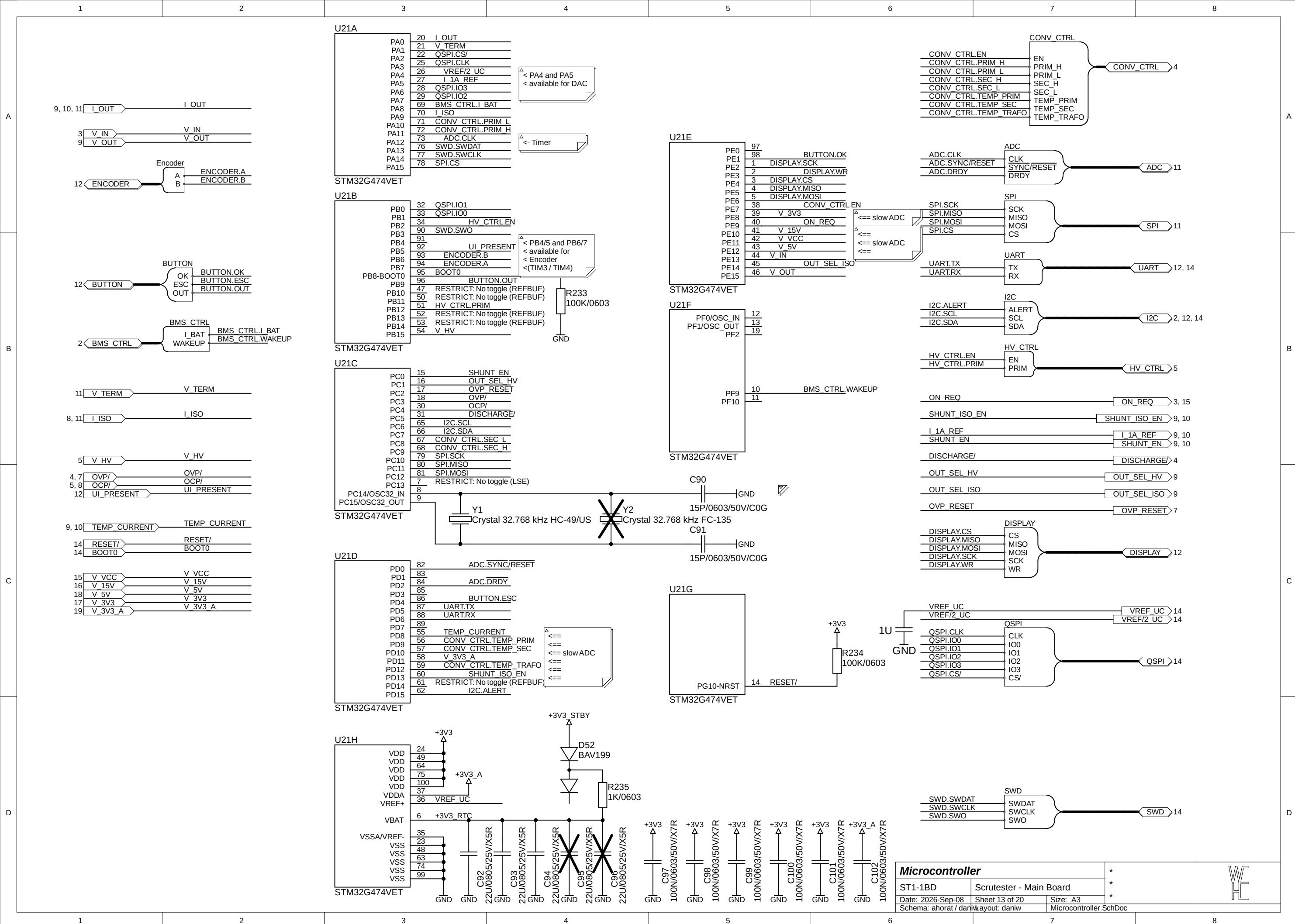


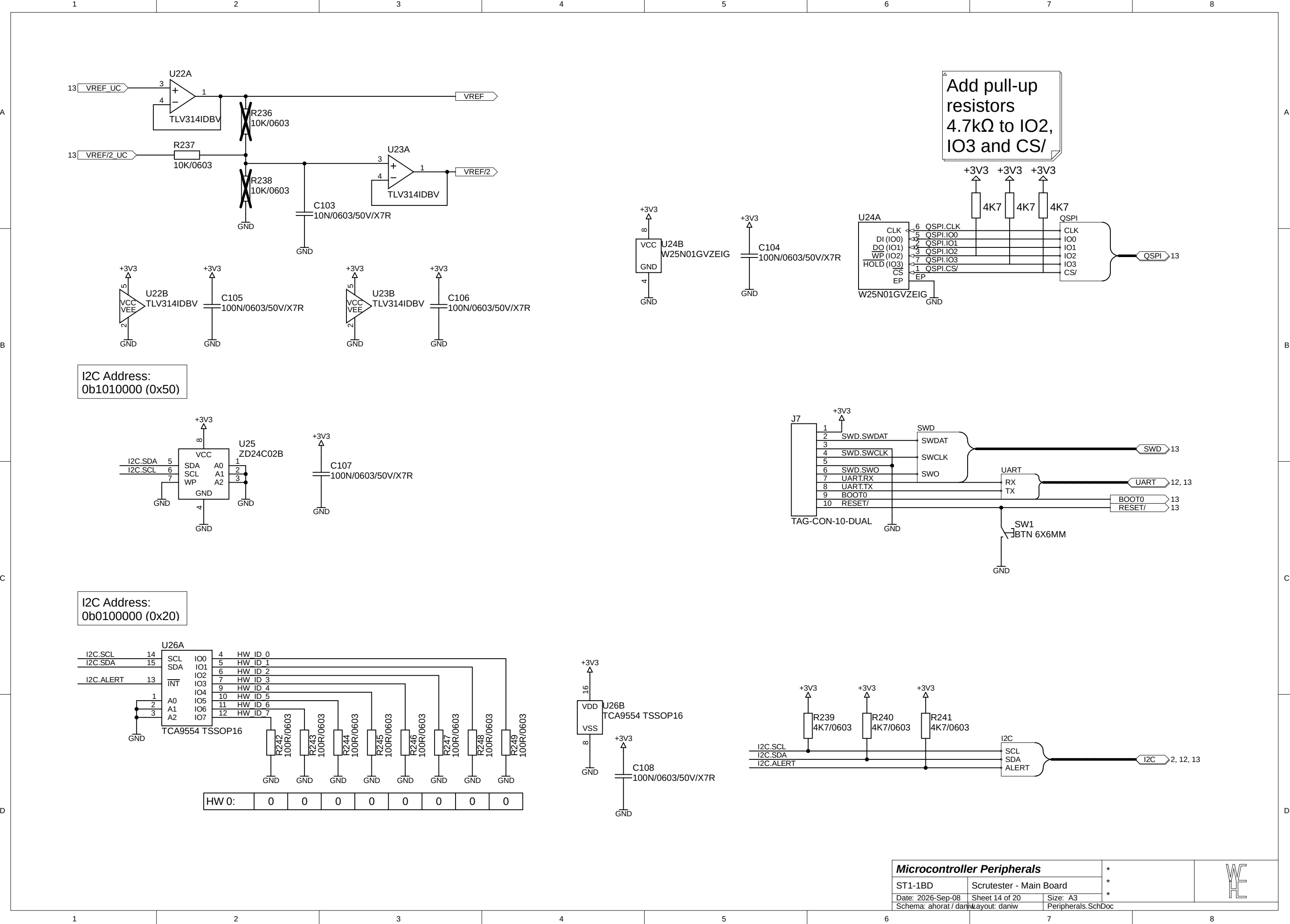


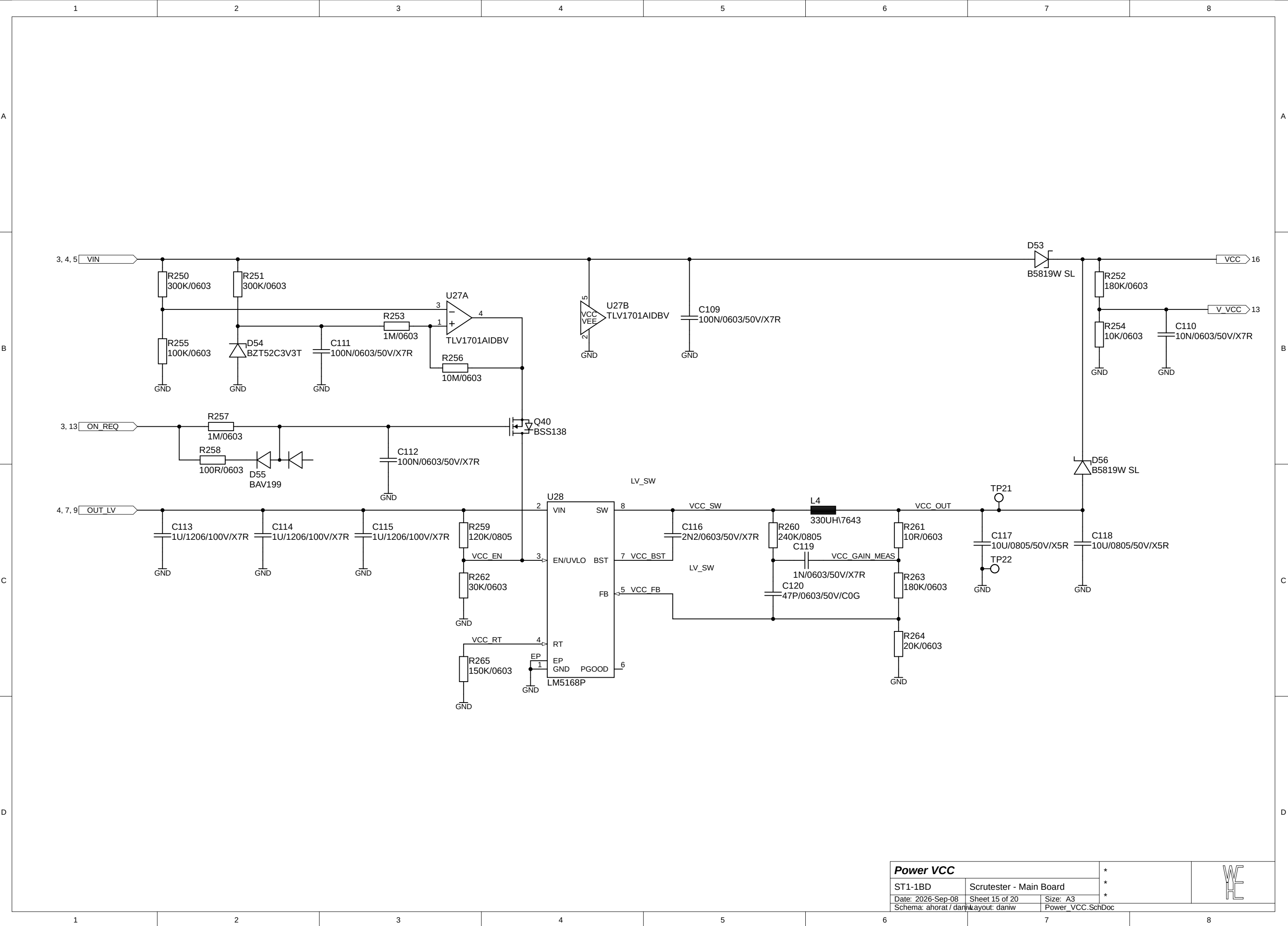


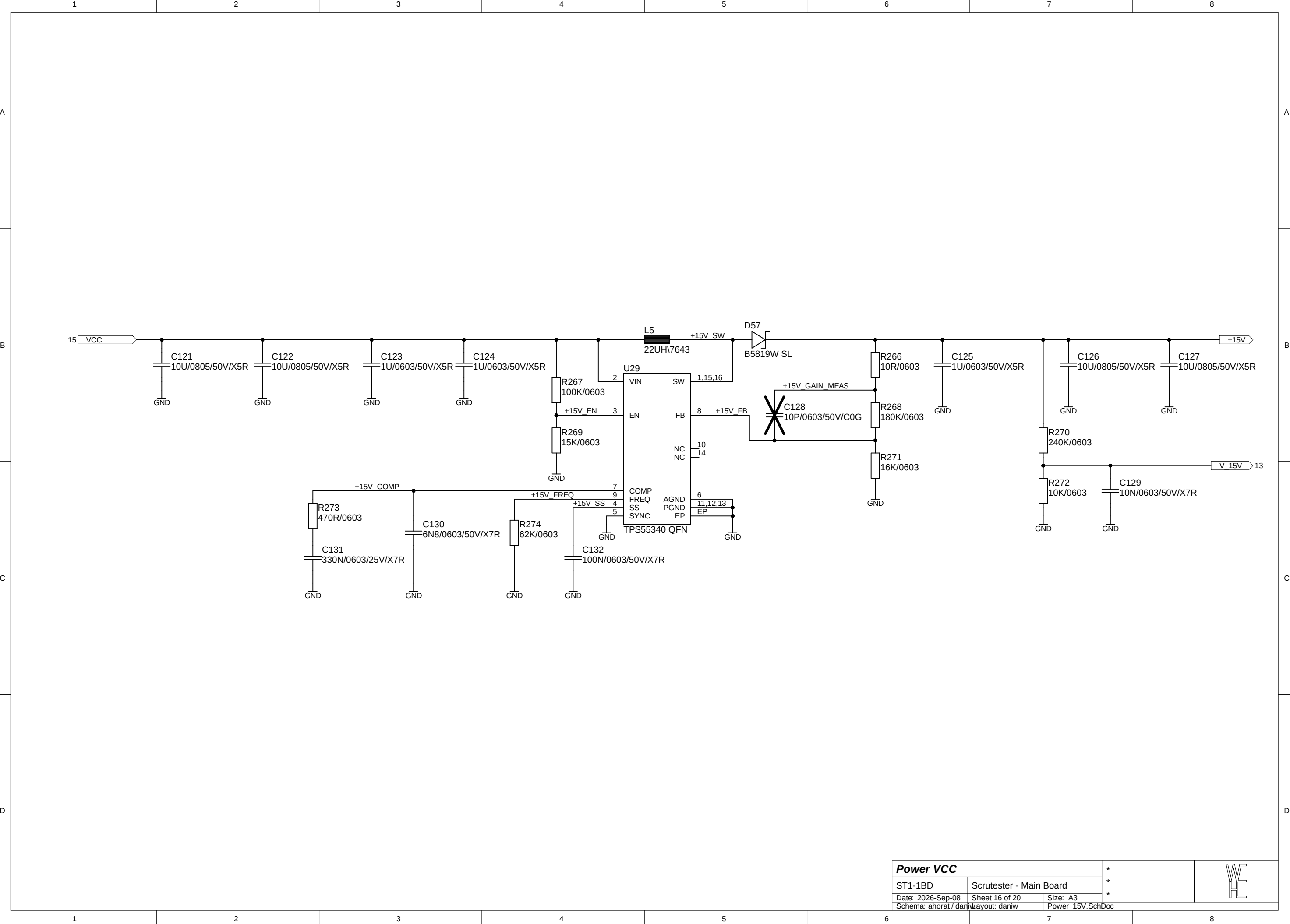
Measurement			*	
ST1-1BD	Scrutester - Main Board		*	
Date: 2026-Sep-08	Sheet 11 of 20	Size: A3	*	
Schema: ahorat / danil	Layout: danil	Measurement.SchDoc		

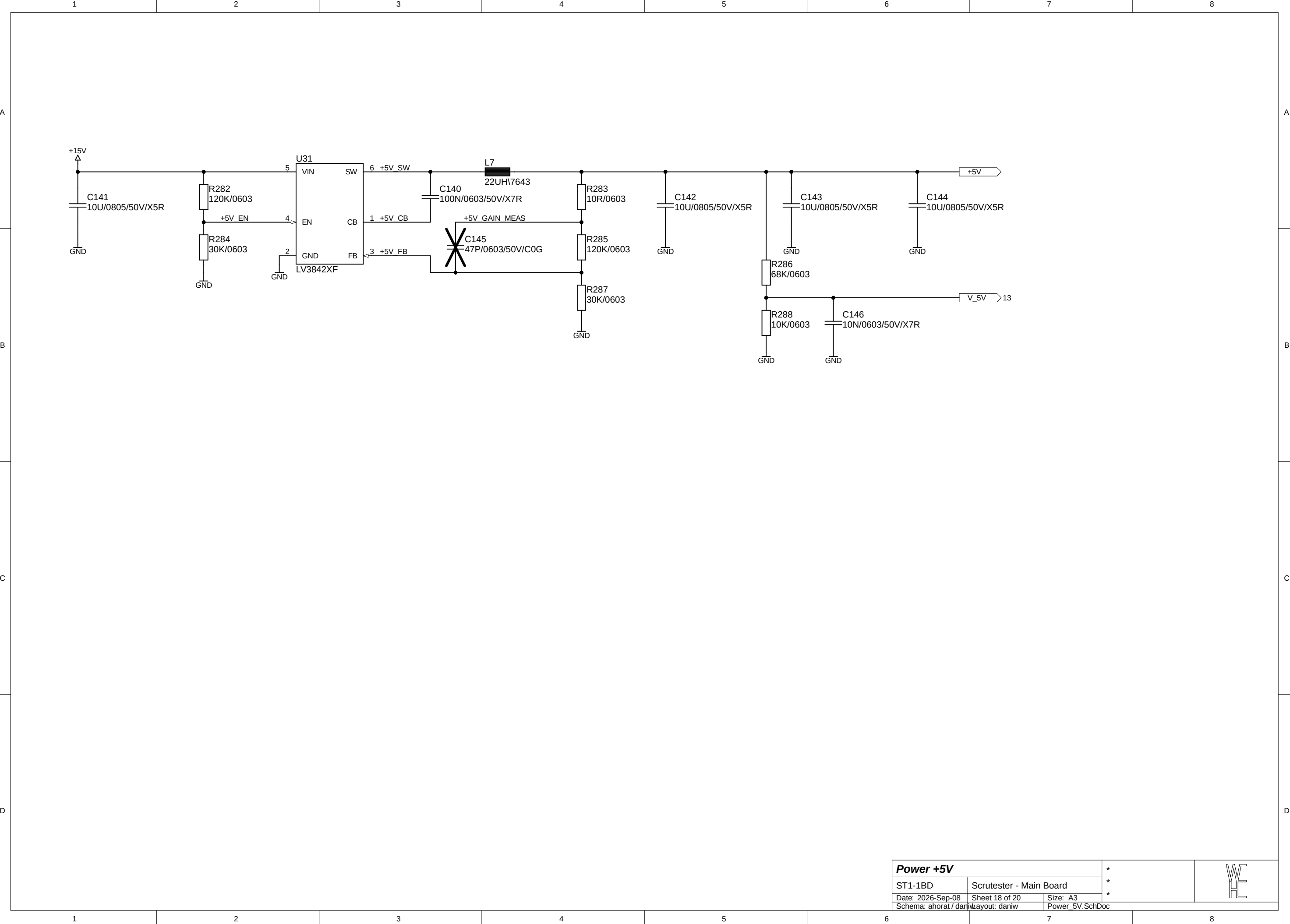


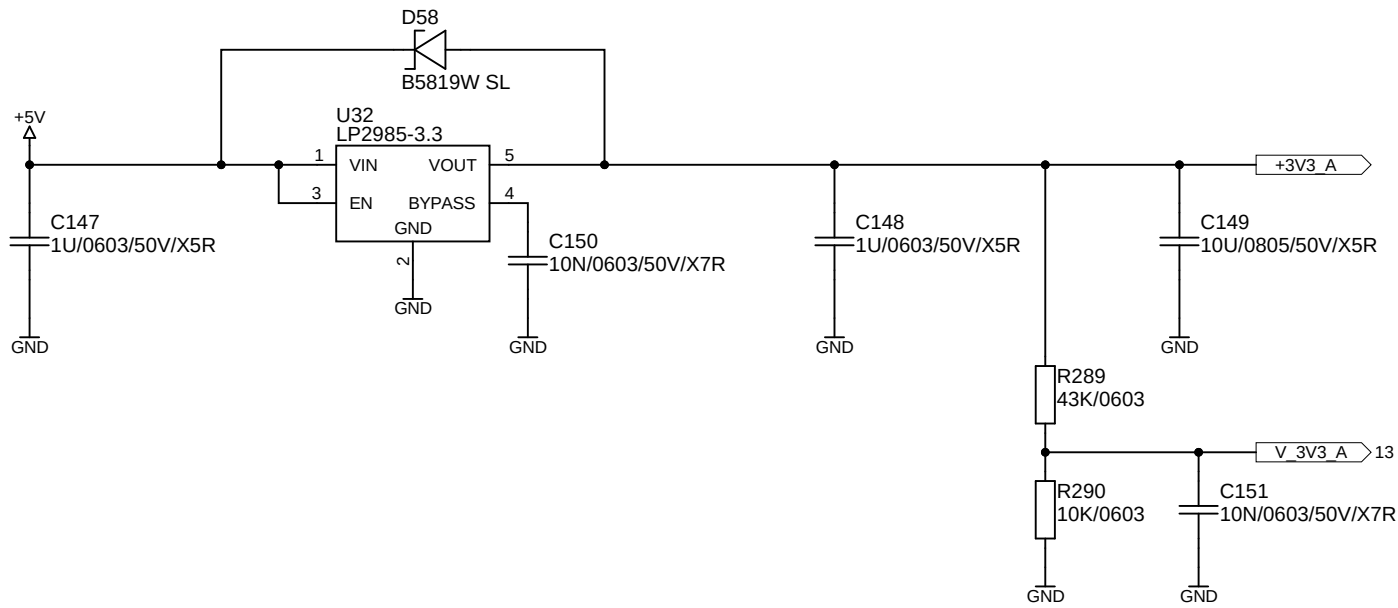












	1	2	3	4	5	6	7	8
A								A
B								B
C								C
D								D
	1	2	3	4	5	6	7	8

Mechanics				
ST1-1BD	Scrutester - Main Board			
Date: 2026-Sep-08	Sheet 20 of 20	Size: A3		
Schema: ahorat / daniw		Layout: daniw		
Mechanics.SchDoc				